Explainable Suicide Risk Detection from Social-Media Text: Fine-Tuned Mental-Health Transformers, Verbatim Evidence, and Factor Identification

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***Abstract*—Suicide-risk assessment from social-media text needs reliable severity classification and an account of why a post received its label. We address the IEEE BigData 2026 Explainable Suicide Risk Detection Challenge: a four-level risk label (Indicator, Ideation, Behavior, Attempt) with verbatim supporting evidence, plus multi-label identification of 24 suicide-related factors. We replace an earlier TF-IDF/logistic-regression pipeline with a fine-tuned MentalRoBERTa encoder, blended with a TF-IDF member for risk, a multi-label MentalRoBERTa model with cross-fitted per-factor thresholds for factors, and an LLM-assisted evidence selector whose outputs are programmatically verified to be exact substrings of the post. Evaluation is user-grouped: an untouched 20% held-out set (330 posts, 32 users) is scored once, after all choices are made by five-fold grouped cross-validation on the remaining 1,305 posts. On the same held-out set, risk accuracy rises from 62.4% to 76.7% (weighted F1 0.621 to 0.767; macro F1 0.563 to 0.751), factor macro F1 from 0.389 to 0.481, and evidence token-F1 from 0.229 to 0.318. The strongest gain is on Behavior and Attempt, and 80.5% of remaining risk errors fall between adjacent severity levels. These are local validation results on a small set, not official leaderboard scores; the evidence metric is a token-overlap proxy, and the LLM step relies on an external API.**

***Index Terms*—Suicide risk detection, natural language processing, explainable AI, evidence extraction, multi-label classification, MentalRoBERTa, group-wise validation, mental-health NLP**

I. Introduction

A. Background and Research Problem

People in distress often describe suicidal thoughts in online communities, and automated analysis of such text has been studied as a screening and triage aid [2], [3]. The task is hard for several reasons: severity is expressed indirectly and depends on context, negation, and intent; adjacent severity levels share much of their vocabulary; social-media text is informal and noisy; labels are imbalanced; and corpora contain several posts per author, so random splitting leaks author-specific style into evaluation.

A bare label gives a human reviewer no basis for checking or contesting the output, which is problematic in a high-stakes domain [4]. The challenge [1] therefore couples classification with two explanatory outputs: the exact phrases that support the risk level and the psychosocial factors present in the post.

B. Contributions

Each contribution is implemented and measured in our notebooks.

1) An explainable three-output pipeline: a four-class risk label, verbatim evidence, and 24-category factor identification.
2) A move from a TF-IDF/logistic-regression pipeline to a fine-tuned MentalRoBERTa encoder [11] for risk and factors, with a measured "without/with" comparison on an identical held-out set.
3) A verified-verbatim evidence selector: an LLM proposes spans, and only spans that match the post text exactly are kept.
4) A stricter validation protocol (nested user-grouped splits, cross-fitted thresholds and blend weight, a held-out set scored once, user-resampled confidence intervals), under which we re-evaluate and correct earlier conclusions.
5) A record of rejected or failed components (class-bias calibration, DeBERTa-v3 run aborted by an error, per-factor threshold tuning for the classical model).

II. Problem Formulation

Let x be a post. The risk model $f_r(x) \to y_r \in \{\text{Indicator}, \text{Ideation}, \text{Behavior}, \text{Attempt}\}$; the evidence model $f_e(x) \to E$, a set of spans, each a substring of x (the annotation "none" is the usual reference for Indicator posts); and the factor model $f_f(x) \to Y_f \subseteq \mathcal{F}$, $|\mathcal{F}| = 24$.

Risk metrics. Weighted F1 averages class F1 by support n_c among N posts, and macro F1 averages classes equally:

$$F1_w = \sum_{c=1}^{4} \frac{n_c}{N} F1_c, \qquad F1_{\text{macro}} = \frac{1}{4} \sum_{c=1}^{4} F1_c. \tag{1}$$

We also report accuracy and quadratic weighted kappa (QWK) [27], which treats the classes as ordered.

Factor metrics. Macro F1 averages the $K{=}24$ binary factor F1 scores; micro F1 pools all decisions; samples F1 averages per-post F1.

Evidence metrics. Our current evaluator is a token-overlap F1 between the predicted evidence string and the concatenated reference spans (bag of word tokens). It does *not* reproduce the challenge's span-matching metric. For the earlier pipeline we also computed a local Phrase F1: a predicted span matches a reference if one contains the other (ignoring case) and its length is at most three times the reference's, with greedy one-to-one matching, and a post with no reference and no prediction scoring 1. Because we lack the organizers' scorer, both are local proxies. A caveat: when the reference is the literal string "none", token-F1 scores the post 0 whether or not the system abstains.

III. Related Work

Mental-health NLP. Language on social media carries signal about depression [5] and suicide risk [2]; Ji *et al.* survey suicidal-ideation detection [3]. Severity formulations include the CLPsych 2019 shared task [6], [7] and knowledge-aware severity assessment [8]. These studies predict labels and do not require span-level evidence.

Pretrained encoders. BERT [9] and RoBERTa [10] are standard fine-tuning backbones; MentalBERT and MentalRoBERTa [11] continue pretraining on mental-health text. DeBERTa-v3 [12] is a stronger general-purpose alternative. We use fine-tuning practices such as discriminative (layer-wise) learning rates [13], decoupled weight decay [14], and label smoothing [15].

Linear baselines. TF-IDF [16] with linear models [17], [19] is a strong, inspectable baseline on small corpora, and character features help on noisy text [18]; this is our "without" system.

Explainability. Post-hoc explainers such as LIME [20] and SHAP [21] attribute importance to features, whereas rationale benchmarks such as ERASER [22] ask whether extracted text supports a prediction; our evidence output follows the rationale view. Large language models can select supporting text from a prompt; we use the open-weight gpt-oss-120b [23] through a hosted API, and verify its output against the source.

Multi-label learning, thresholds, validation. One-vs-rest decomposition [24], per-class thresholds for F1 [25], and group-wise validation [26] are standard; we apply thresholds cross-fitted on out-of-fold predictions.

IV. Dataset and Data Analysis

A. Data and Annotations

The training file has 1,635 labeled posts (columns: `row_id`, `anon_user_id`, `post_id`, `post`, risk label, evidence, factors) from 153 anonymized users (7–30 posts per user, median 9). A separate file of 502 unlabeled posts from 48 users, disjoint from the training users, is the prediction set; it is not scored in this paper. Evidence is a semicolon-separated list of phrases and factors a string-encoded list.

B. Risk Labels and Imbalance

After normalizing ten raw label spellings, the classes are Indicator 611 (37.4%), Ideation 519 (31.7%), Behavior 391 (23.9%), and Attempt 114 (7.0%) (Fig. 1).

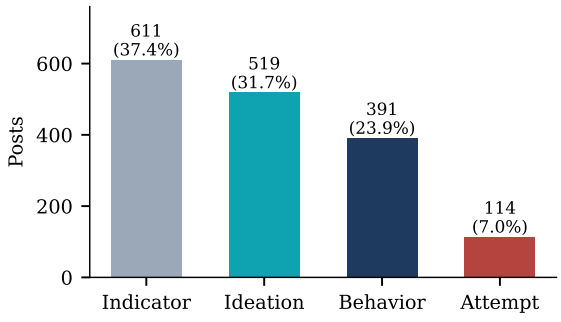


Fig. 1. Risk-level distribution of the 1,635 training posts.

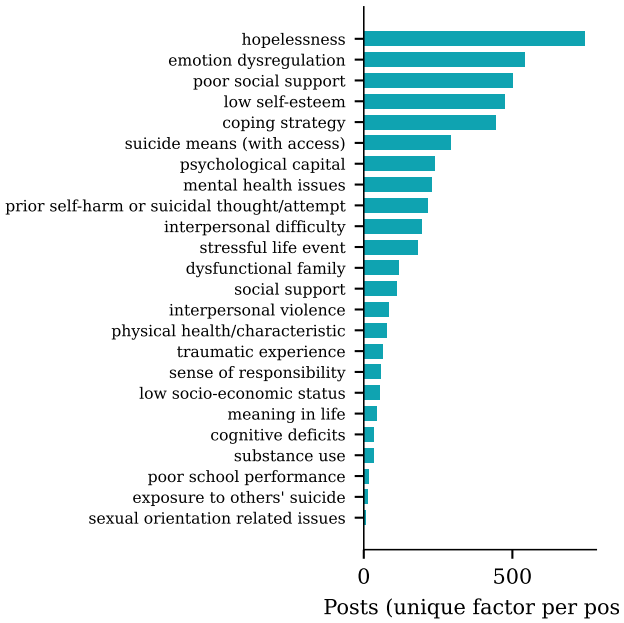


Fig. 2. Frequency of the 24 factor categories (unique factor per post).

C. Evidence, Factors, and Length

Of the training posts, 583 (35.7%; 95.4% of Indicator posts) are annotated "none", and the other 1,047 labeled posts carry 1,833 phrases (1.75 per post) with a median length of 4 tokens (90th percentile 8); five Indicator posts have no evidence field. The 24 factors follow a long-tailed distribution (Fig. 2): hopelessness appears in 745 posts, sexual-orientation-related issues in 8. Factor lists repeat labels in 812 posts (deduplicated), 128 posts have no factor, and the mean is 2.92 unique factors per post. Posts are short with a heavy tail: median 47 and 95th-percentile 279 MentalRoBERTa tokens, with 1.7% exceeding 510.

D. Data Quality

After cleaning, 85 rows share their text with another row, and two such duplicated texts carry conflicting risk labels; one exact text occurs in both the development and held-out

sets (we kept it; its effect on a 330-post set is at most one post).

V. Preprocessing and Validation Protocol

A. Preprocessing

We unescape HTML, strip HTML tags, remove URLs, mask user and subreddit mentions (`[USER]`, `[SUBREDDIT]`), and collapse whitespace. We keep case, punctuation, and negations, because words such as "not" or "never" can reverse meaning, and transformers benefit from natural text. Long posts are truncated to 512 tokens keeping the first half and last half, rather than losing the tail.

B. Nested User-Grouped Evaluation

All splits use `StratifiedGroupKFold` on `anon_user_id`, so no user appears on both sides of any split. A first split sets aside a 20% *held-out* set (330 posts, 32 users: Indicator 113, Ideation 117, Behavior 80, Attempt 20) and leaves a development set (1,305 posts, 121 users). Five-fold grouped cross-validation on the development set produces out-of-fold (OOF) predictions for every development post. All choices (blend weight, thresholds, whether to use class biases) are made on OOF predictions, and the held-out set is scored once, at the end. We report 95% confidence intervals from 2,000 bootstrap resamples of *users*. The single seed is 42; unlike our earlier multi-seed protocol (Section VI-D), results here come from one split. OOF scores are slightly optimistic because each fold's best epoch is chosen on that fold; we therefore quote held-out numbers.

VI. Methodology

Fig. 3 summarizes the system.

A. Risk Classification

Encoder. `mental/mental-roberta-base` [11] with masked mean pooling, dropout 0.2, and a linear head, fine-tuned on each of the five development folds. Settings: learning rate 3×10^{-5} with layer-wise decay 0.9, head learning rate 2×10^{-4}, AdamW with weight decay 0.01, 10% linear warm-up, batch 8 with 2-step gradient accumulation, at most 5 epochs with patience 2, label smoothing 0.05, class weights proportional to (inverse frequency)$^{0.5}$, FP16 mixed precision. The held-out prediction is the average of the five fold models. **TF-IDF member.** Word 1–2-gram (sublinear, up to 150,000 features) and character-wb 3–5-gram (up to 200,000 features) TF-IDF with logistic regression (C=3). **Blend.** Probabilities are mixed as $(1-w)P_{\text{enc}} + wP_{\text{tfidf}}$ with w=0.20, chosen by OOF log-loss over $\{0, 0.05, \ldots, 0.5\}$. Four additive class biases fitted on OOF to maximize accuracy were evaluated cross-fitted and *rejected* because they did not help. A DeBERTa-v3 [12] member was configured, but its run stopped with an FP16 gradient-unscaling error; no result exists and it is not part of the system.

B. Factor Identification

The same MentalRoBERTa encoder is fine-tuned with 24 sigmoid outputs and positive-class weights capped at 10. Each factor k has its own threshold t_k chosen on OOF probabilities over $\{0.10, 0.12, \ldots, 0.90\}$ to maximize that factor's F1; factors with fewer than 8 positives keep t_k=0.5. OOF scores use thresholds tuned on the other four folds (cross-fitted), and the held-out set uses thresholds tuned on all OOF predictions, so no threshold sees held-out labels.

C. Evidence Selection

Evidence (Fig. 4) is selected by gpt-oss-120b via a hosted API (temperature 0, low reasoning effort, three parallel workers), given the post, the predicted risk level and a short definition of it, and asked for one to three spans of at most 25 words, copied exactly. Each returned span is searched in the post after folding curly quotes, dashes, case and whitespace, and the matched *original* characters are returned; spans without a match are dropped. If nothing survives, the sentence with the most risk-cue matches is used. Hence every output is a substring of the post, although the choice of span is made by a model we do not control and which can change. On the held-out set 94.85% of posts used an LLM span and the rest the fallback (17 of 330; the log shows rate-limit errors, which we did not verify case by case). The selector never abstains, so it returns text even for Indicator posts whose reference is "none".

D. Earlier Classical Pipeline ("Without")

The earlier system used: word 1–2-gram plus character 3–5-gram TF-IDF with class-balanced logistic regression (C=1) for risk; word TF-IDF with one-vs-rest logistic regression for factors with per-factor thresholds; and a phrase-level span scorer (sentences and sub-clauses scored by TF-IDF logistic regression, threshold 0.7, shortest-span preference) for evidence. On its own, earlier user-grouped split (380 validation posts) it reached risk weighted F1 0.6468 (5-seed mean 0.6356; word-only 0.6215, with character features better on all five seeds), local Phrase F1 0.5599 (3-seed mean 0.5246), and factor macro F1 0.4419 (3-seed mean 0.4235). That split differs from the one used here, so those figures are *not* comparable with the current ones; for the comparison below we re-ran the earlier pipeline on the current split.

VII. Experimental Setup

Fine-tuning ran on a Tesla T4 GPU (Colab) with PyTorch and Transformers; each fold took 88–109 s for risk and 87–93 s for factors. Classical baselines were run with scikit-learn [19], using the split algorithm version that reproduces the notebook's held-out class counts exactly; as a check, our re-implementation of the notebook's TF-IDF member reproduced its OOF accuracy (0.6169), OOF macro F1 (0.4908), and held-out accuracy (0.6273) to the reported digits. The earlier pipeline was re-run on the same development/held-out split and folds, with factor thresholds tuned under the current cross-fitted protocol.

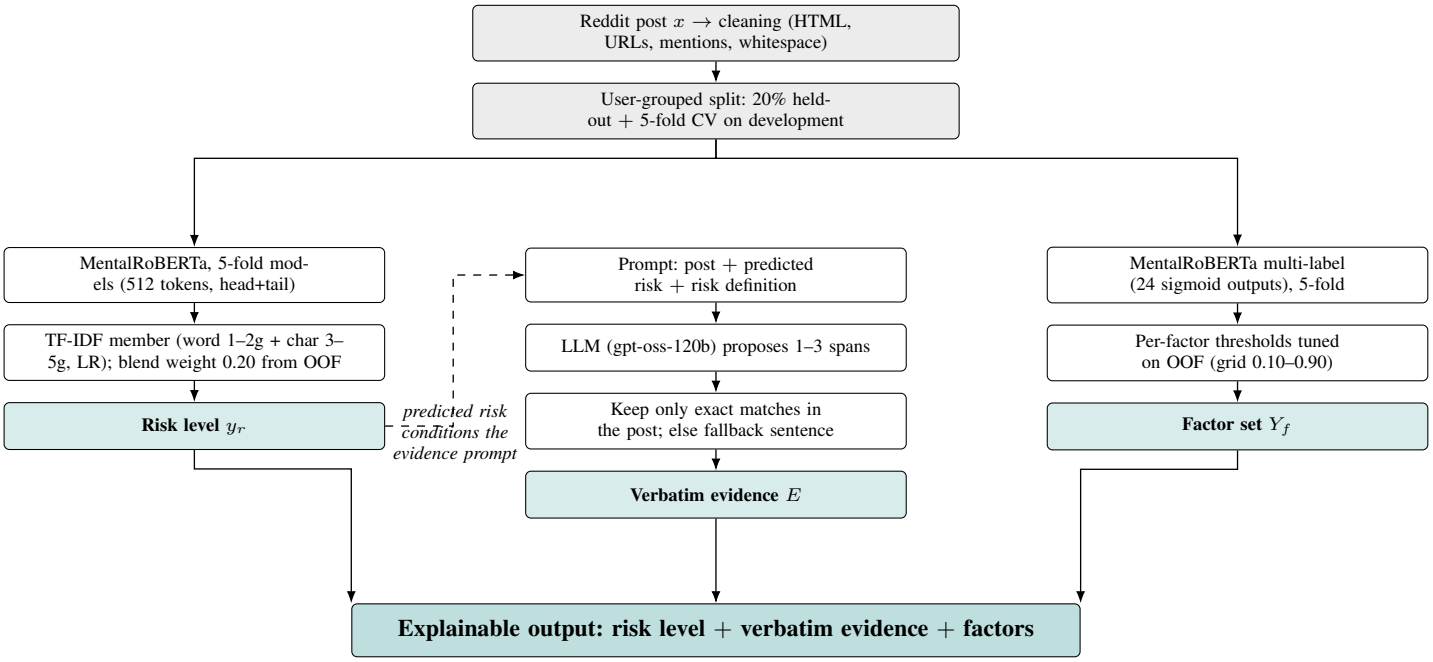


Fig. 3. Architecture of the current framework. The post is cleaned and split by user; risk comes from a MentalRoBERTa/TF-IDF blend, factors from a multi-label MentalRoBERTa model, and evidence from an LLM whose spans are verified against the source.

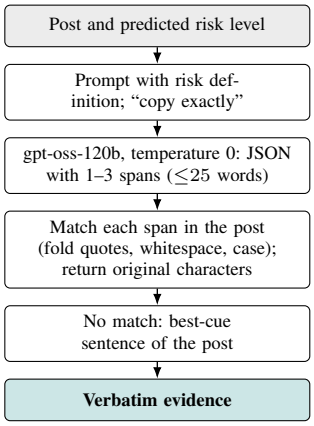


Fig. 4. Evidence selection. Whatever the LLM returns, the output text is copied from the post.

VIII. Results and Analysis

A. Without vs. With

Table I and Fig. 5 show gains on every metric. For risk, held-out accuracy is 76.67% (95% CI 72.06–80.93) with the current system and 62.42% (95% CI 55.14–69.52) for the earlier one; the intervals, computed separately by user resampling, do not overlap, but with 32 users they are wide, and we did not compute a paired test. On the development set (OOF) the current risk model reaches accuracy 75.48% (95% CI 71.98–78.87), macro F1 71.36%, weighted F1 75.47%, balanced accuracy 70.85%, and QWK 0.764; its per-fold accuracy is 75.50% $\pm$ 0.65%.

TABLE I
Held-Out Results (330 Posts, 32 Users): Without vs. With

Task	Metric	Without	With
Risk	Accuracy	0.6242	**0.7667**
Risk	Macro F1	0.5631	**0.7513**
Risk	Weighted F1	0.6209	**0.7671**
Risk	QWK	0.5626	**0.7560**
Factors	Micro F1	0.5465	**0.6315**
Factors	Macro F1	0.3888	**0.4811**
Factors	Samples F1	0.3972	**0.5252**
Evidence	Token F1	0.2291	**0.3180**

Without: earlier TF-IDF/linear pipeline re-run on this split. With: MentalRoBERTa+TF-IDF (risk), MentalRoBERTa (factors), LLM evidence with verification. Local validation scores, not official results.

TABLE II
Per-Class F1 on the Held-Out Set

Class	Support	Without	With	With (P / R)
Indicator	113	0.737	0.820	0.835 / 0.805
Ideation	117	0.658	0.748	0.736 / 0.761
Behavior	80	0.457	0.738	0.738 / 0.738
Attempt	20	0.400	0.700	0.700 / 0.700

B. Per-Class Risk Results

Table II shows the largest gains on Behavior (+0.28) and Attempt (+0.30). Attempt has only 20 held-out posts, so its figure moves by about 0.05 for every misclassified post.

C. Stage-by-Stage Ablation

Table III attributes the gain to the encoder: replacing TF-IDF by MentalRoBERTa raises OOF accuracy by about 14 points. The TF-IDF blend, chosen by OOF log-loss, changes accuracy

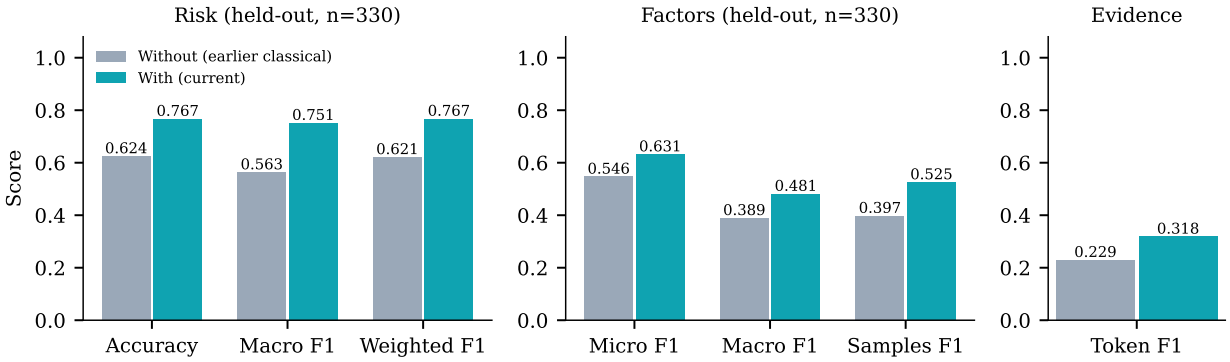


Fig. 5. Held-out comparison of the earlier classical pipeline (without) and the current system (with) on identical data.

TABLE III
RISK ABLATION (DEVELOPMENT OOF AND HELD-OUT)

Stage	OOF acc.	OOF macro F1	Held-out acc.
Earlier TF-IDF, word only	0.5946	0.5290	0.6061
Earlier TF-IDF, word + char	0.6169	0.5537	0.6242
Current TF-IDF member	0.6169	0.4908	0.6273
MentalRoBERTa (5 folds)	0.7571	0.7218	0.7606
+ TF-IDF blend (w=0.20)	0.7548	0.7136	0.7667
+ class biases	rejected (did not help)		

TABLE IV
FACTOR IDENTIFICATION (MACRO / MICRO F1)

	OOF		Held-out	
Model	Macro	Micro	Macro	Micro
Classical, fixed 0.5	0.358	0.566	0.391	0.574
Classical, tuned t_k	0.362	0.514	0.389	0.547
MentalRoBERTa, tuned t_k	**0.423**	**0.585**	**0.481**	**0.632**

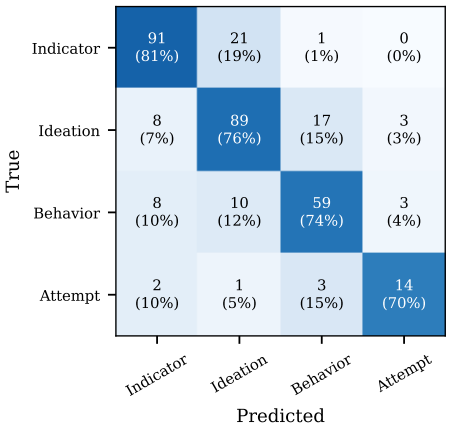


Fig. 6. Held-out confusion matrix of the current risk model (counts and row percentages).

by -0.0023 on OOF and $+0.0061$ on held-out, which is within noise for 330 posts; its contribution is not demonstrated. Several settings (512-token head+tail truncation, class-weight power, layer-wise decay, label smoothing) were changed together and were not ablated individually.

D. Factor Identification

Table IV reports factors under the current protocol. Two findings follow. First, MentalRoBERTa improves macro F1 over the classical models by about 0.06 on OOF and 0.09 on the held-out set. Second, per-factor tuning gave the classical model no gain once thresholds were cross-fitted (macro F1 0.358 to 0.362 OOF, 0.391 to 0.389 held-out) and lowered micro F1; the earlier $+0.056$ gain came from tuning on the evaluation split itself. This corrects the earlier conclusion. We do not have an untuned MentalRoBERTa score, so the benefit of tuning for the transformer is not isolated. Per-factor results are weakest for rare categories (OOF F1: sexual-orientation-related issues 0.00 with 6 positives, cognitive deficits 0.09, exposure to others' suicide 0.10, meaning in life 0.15, physical health 0.20, substance use 0.20).

E. Evidence Extraction

On the held-out set, the LLM-assisted selector reaches token F1 0.318 with 100% coverage (94.85% from the LLM), against 0.229 and 54.8% coverage for the earlier phrase-level extractor under the same metric. In 110 of the 330 posts the reference is "none" and both systems score 0 on them, so restricting to the 220 posts with real references gives 0.344 for the earlier extractor and, under that same assumption, about 0.48 for the LLM selector (a derived estimate, $0.318 \times 330/220$). We could not compute a Phrase F1 for the LLM selector because its predictions are not stored in the notebook, and because token-F1 scores "none" posts 0 regardless of abstention, it does not penalize the LLM selector for emitting text on Indicator posts, where an abstention-aware metric would. The comparison therefore probably overstates the LLM's advantage.

F. Confusion Matrix and Risk Errors

Fig. 6 shows that 80.5% of errors are between adjacent severity levels (80.6% on OOF). The largest held-out confusion is Indicator predicted as Ideation (21 posts, 18.6% of Indicator), followed by Ideation to Behavior (17)

and Behavior to Ideation (10). On OOF the leading confusions are Indicator→Ideation (62), Behavior→Ideation (56), Ideation→Behavior (48) and Ideation→Indicator (39). Severity levels differ by degree of action rather than topic, which limits separability; the earlier classical model made 124 errors on the held-out set, 31 of them (25%) between non-adjacent levels, against 15 of 77 (19.5%) for the current model.

IX. Explainability and Evidence Analysis

For each post the system returns (1) a risk level, (2) verbatim supporting text, and (3) factor categories. A held-out example from the notebook output: for the post "so ugly that i want to kill myself because of it", the model predicts Ideation (probability 0.94, the annotated label), factor low self-esteem (annotated), and selects the whole post as evidence, where the reference is the shorter phrase "want to kill myself". This illustrates the strength (checkable text, consistent factor) and a recurring weakness (granularity). We call this *model-supported textual evidence*; it is not clinical validation. Because the span selector and the risk classifier are separate models, the evidence need not reflect what drove the risk prediction. The earlier verified-verbatim property holds in the new pipeline by construction of the matching step.

X. Error Analysis

Risk. Errors cluster at neighboring levels, with the weakest class being Attempt (OOF F1 0.59; held-out 0.70 on 20 posts) and the largest single confusion Indicator→Ideation. **Factors.** Rare and abstract categories are weak and over-predicted: OOF predictions exceed supports for exposure to others' suicide (45 predicted for 13 true), substance use (95 for 33), meaning in life (68 for 36), and cognitive deficits (38 for 27). The classical model over-predicts overall (held-out 3.7 factors per post against 2.8 annotated with tuned thresholds); we did not compute this statistic for the transformer. **Evidence.** The notebook stores no per-post evidence predictions, so we do not give a span-level error breakdown; the token-F1 of 0.318 and the example above indicate that granularity and the "none" convention are the main open problems.

XI. Responsible AI, Ethics, and Limitations

Research prototype. This system is a research prototype and is not a clinical diagnostic system. It should support trained human assessment rather than replace professional judgment, and we make no claim about real-world suicide-prevention effectiveness.

Risks. False negatives may leave a person in need unflagged; false positives may trigger unwarranted intervention. Our held-out accuracy of 76.7% means roughly one post in four is misclassified. Models trained on one community may transfer poorly across cultures and platforms, and we have not audited demographic bias. **Data governance.** The evidence step sends post text to a third-party hosted LLM. For sensitive mental-health data this requires explicit approval under the data-use terms and should be replaced by a locally hosted model in any real deployment; reproducibility also depends on the provider's model remaining available.

Limitations as research boundaries. (i) 1,635 posts from 153 users, and a held-out set of 330 posts from 32 users, so confidence intervals are about ± 4.4 points; Attempt has 20 held-out posts. (ii) A single split seed in the current protocol; the earlier multi-seed checks were not repeated for the transformer. (iii) One DeBERTa-v3 run failed and no second backbone was evaluated. (iv) The evidence metric is a token proxy and the LLM's Phrase F1 was not measured. (v) Several training settings were changed jointly. (vi) Local validation need not match the official leaderboard.

XII. Discussion

Why the encoder helps. A mental-health-adapted encoder uses context, negation and word order, which bag-of-words features cannot, and it matters most for classes distinguished by action (Behavior, Attempt). **Why the protocol matters.** Under stricter evaluation, a classical improvement we had celebrated (per-factor thresholds, $+0.056$) vanished, and a blend that looked useful was within noise; reporting a held-out set scored once with user-resampled intervals changes which claims survive. **Trade-offs.** The transformer gains about 14 accuracy points at the cost of GPU training, a less inspectable model, and (for evidence) dependence on an external LLM; the classical pipeline remains a fully local, auditable fallback. **What explainability adds.** Verified-verbatim evidence makes outputs checkable, but its usefulness is uneven and its fidelity to the classifier is not established.

XIII. Conclusion and Future Work

Replacing a TF-IDF/linear pipeline with fine-tuned MentalRoBERTa models and an LLM-assisted, verified evidence selector raised held-out risk accuracy from 62.4% to 76.7% (weighted F1 0.621 to 0.767), factor macro F1 from 0.389 to 0.481, and evidence token-F1 from 0.229 to 0.318 on the same 330 posts. Remaining challenges are rare factors, the Indicator/Ideation boundary, evidence granularity, and the evidence step's external dependency.

Future work: multiple seeds and backbones (including a repaired DeBERTa-v3 run) with significance testing; ablating each fine-tuning setting; storing evidence predictions to compute the challenge's span metric and enabling abstention on "none" posts; a locally hosted open model for evidence; calibration and uncertainty estimation; fairness and cross-domain evaluation; and human-in-the-loop studies with clinicians. None of these has been done here.

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